

PHYS205 Experiment 3 Lab Report: Poiseuille's Law for Flow of Fluids

Ahmet Buğra Ertürk

Lab Partner(s): Egemen Saritekin, Berkay Furkan Çakmak

November 26, 2025

1 Objective

The aim of this experiment is to explore the dependence of the flow rate, during laminar flow, on the viscosity of the fluid, pressure drop at both ends of the pipe and the pipe dimensions (radius and length).

2 Theory

Viscosity

Viscosity is internal friction in a fluid. Viscous forces oppose the motion of one portion of a fluid relative to another. Viscosity is the reason it takes effort to paddle a canoe through calm water, but it is also the reason the paddle works.

Viscosity has important effects on the flow of liquids through pipes, including blood flow in the circulatory system. Figure below helps visualize the flow-speed profile for laminar flow of a viscous fluid in a long cylindrical pipe. Due to viscosity, the speed is zero at the pipe walls and greatest at the center of the pipe.

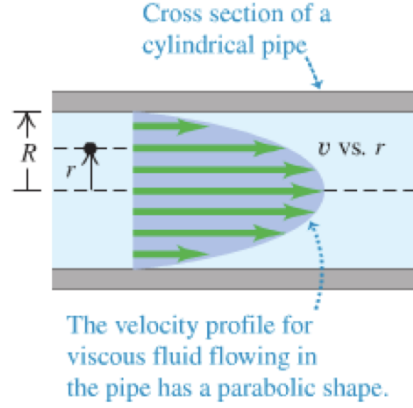


Figure 1: Velocity profile for a viscous fluid in a cylindrical pipe.

Poiseuille's Law

For a viscous fluid, the rate of laminar flow through a cylindrical tube of Radius R and length L is given by Poiseuille's Law:

$$Q = \frac{\pi R^4 (P_1 - P_2)}{8\eta L} \text{cm}^3/\text{sec} \quad (1)$$

Where $P_1 - P_2$ is the pressure difference between the two ends of the cylinder, η is the coefficient of viscosity measured in $\text{dyn}^1(\text{sec}/\text{cm}^2)$.

The pressure difference required to sustain a given volume flow rate through a cylindrical pipe of length L and radius R is proportional to L/R^4 . If we decrease R by one-half, the required pressure increases by 16. This simple relationship explains the narrowing arteries and high blood pressure due to R^4 dependence, very small narrowing of the arteries results in a substantially elevated blood pressure and added strain on the heart muscle.

The volume flow rate can be also be expressed in terms of cross section of the pipe, A and the velocity of the flow v as;

$$Q = Av \quad (2)$$

¹1 $\text{dyn} = 10^{-5} \text{ N}$

Table 1: Dynamic viscosity of water at different temperatures is listed in SI units.

Temperature t ($^{\circ}C$)	Dynamic Viscosity μ ($\text{dyn} \cdot \text{s}/\text{cm}^2$) $\times 10^{-2}$
0	1.787
5	1.519
10	1.307
20	1.002
30	0.798
40	0.653
50	0.547
60	0.467
70	0.404
80	0.355
90	0.315
100	0.282

Laminar Flow

If the velocity of a fluid exceeds a critical point, the flow becomes turbulent with whirls disrupting the laminar flow. In a cylindrical pipe, this critical velocity is given by

$$v_c = \frac{\text{Re}\eta}{\rho D} \quad (3)$$

Where Re is the Reynold's number, D is the diameter of the cylinder, ρ is the density of the fluid, η is the viscosity. When the Reynold's number is below 2300, the flow is *laminar*, between 2300-4000 the flow is *transient*, and above 4000 the flow is *turbulent*.

3 Procedure

Equipment

- Water Container
- Capillary Tubes

- Thermometer
- Timer
- Clamp Stand
- Level
- Beaker
- Ruler
- Caliper

Experiment

1. Fill the container with water up to 10 cm above the top of the water tap.
2. Indicate the level of water and make sure that it stays the same throughout the experiment.
3. Observe the temperature of the room which corresponds to water temperature.
4. Take a tube, measure its Radius and length and record the values in the table below.
5. Carefully insert the tube through the stopcock and insert the stopcock to the water tap.
6. Measure the height at both ends of the tube to make sure the tube is aligned horizontally and secure the tube with the clamp stand.
7. Place a beaker under the tube opening.
8. Open the water tap and start the timer simultaneously.
9. Wait 30 seconds, then stop the timer and close the tap simultaneously.
10. Measure the collected water in beaker with the help of a falcon tube and record the flow rate in the table below.

11. Disconnect the stopcock from the tap and then disconnect the tube from the stopcock.
12. Repeat the same procedure for the remaining tubes and record the values in the data table.

4 Data

$$Temperature = 22^{\circ}C \quad Height(h) = 10 \text{ cm}$$

Table 2: Flow Rate (Q)

Q (cm ³ /sec)	L ₁ = 10 cm	L ₂ = 20 cm	L ₃ = 30 cm	L ₄ = 40 cm
R ₁ = 0.2 cm	×	×	8.8	×
R ₂ = 0.15 cm	×	×	4.4	×
R ₃ = 0.1 cm	4.57	3.77	3	2.87
R ₄ = 0.05 cm	×	×	0.65	×

5 Data Analysis

1. To calculate the Re, the equations (2) and (3) can be used.

$$Re = \frac{\rho D v_c}{\eta} = \frac{\rho D Q}{\eta A} = \frac{2 \rho Q}{\eta \pi R} \quad (4)$$

Now, plug the our measured values in the Equation (4).

$$Re = \frac{2 \cdot 1 \text{ (g/cm}^3\text{)} \cdot Q \text{ (cm}^3\text{/s)}}{0.9532 \text{ (dyn} \cdot \text{s/cm}^2\text{)} \times 10^{-2} \cdot \pi \cdot R \text{ (cm)}} = \frac{2 \cdot Q}{0.9532 \cdot 10^{-2} \cdot \pi \cdot R} \quad (5)$$

Table 3: Reynold Numbers

Tube	R (cm)	L (cm)	Q (cm³/s)	Re
R_1, L_3	0.2	30	8.8	2939
R_2, L_3	0.15	30	4.4	1959
R_3, L_3	0.1	30	3	2004
R_4, L_3	0.05	30	0.65	868
R_3, L_1	0.1	10	4.57	3052
R_3, L_2	0.1	20	3.77	2518
R_3, L_4	0.1	40	2.87	1917

As mentioned before, Poiseuille's law is valid only for laminar flow ($Re \leq 2300$) so four of our values satisfy this law.

2. To calculate the viscosity with Poiseuille's law, the equations (1) will be used.

$$P_1 = \rho gh = 1 \text{ (g/cm}^3\text{)} \cdot 981 \text{ (cm/s}^2\text{)} \cdot 10 \text{ (cm)} + P_{atm} = 9810 \text{ g/s}^2 + P_{atm} \quad (6)$$

$$P_2 = P_{atm} \quad (7)$$

So, $P_1 - P_2 = 9810 \text{ g/s}^2$.

$$\eta = \frac{\pi R^4 \text{ (cm}^4\text{)} \cdot 9810 \text{ (g/s}^2\text{)}}{8Q \text{ (cm}^3\text{/s)}L \text{ (cm)}} \quad (8)$$

When Poiseuille's law is applied for our eligible values:

Table 4: Poiseuille's Law

Tube	R (cm)	L (cm)	Q (cm³/s)	$\eta \text{ (dyn} \cdot \text{s/cm}^2\text{)}$
R_2, L_3	0.15	30	4.4	1.48×10^{-2}
R_3, L_3	0.1	30	3	0.43×10^{-2}
R_4, L_3	0.05	30	0.65	0.12×10^{-2}
R_3, L_4	0.1	40	2.87	0.34×10^{-2}

$$\eta_{avg} = \frac{1.48 + 0.43 + 0.12 + 0.34}{4} \times 10^{-2} = 0.59 \times 10^{-2} \text{ dyn} \cdot \text{s/cm}^2 \quad (9)$$

3. The theoretical viscosity for 22°C is $\eta = 0.9832 \cdot 10^{-2} \text{ dyn}\cdot\text{s}/\text{cm}^3$

$$\%Error = \frac{|\eta - \eta_{avg}|}{\eta} \times 100 \quad (10)$$

$$\%Error = \frac{|0.9832 \cdot 10^{-2} - 0.59 \cdot 10^{-2}|}{0.9832 \cdot 10^{-2}} \times 100 = 40\% \quad (11)$$

The error is a bit high because our experimental setup has some problems. Water was leaking from our container, also our measurement methods are not precise. Therefore, it is expected even if it is a bit high.

4. For L=30 cm

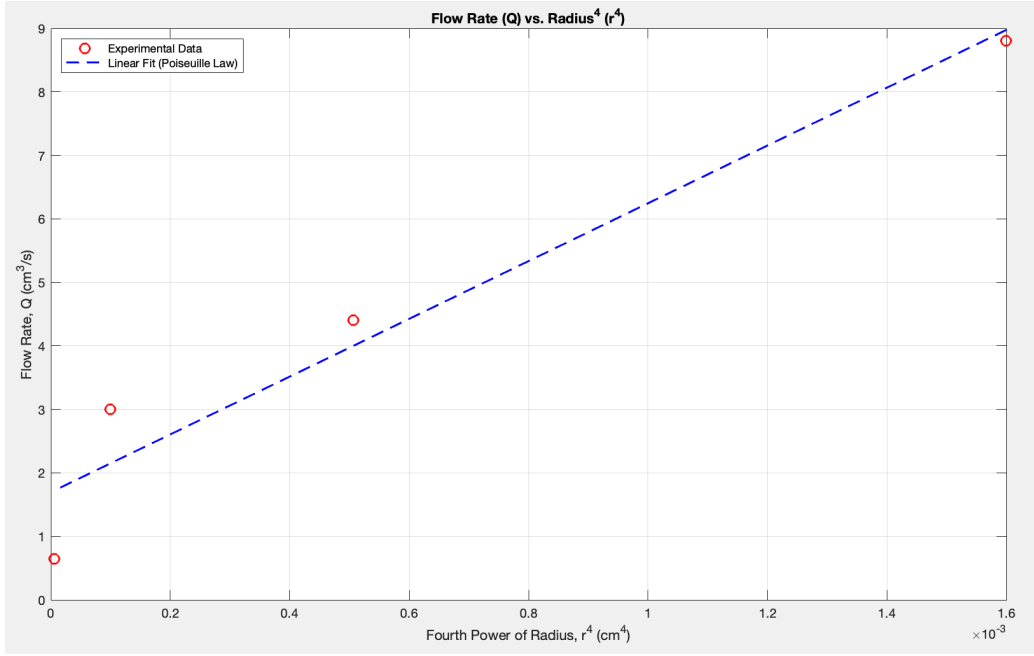


Figure 2: Flow Rate vs. Fourth Power of the Tube Radius

5. For R=0.1 cm

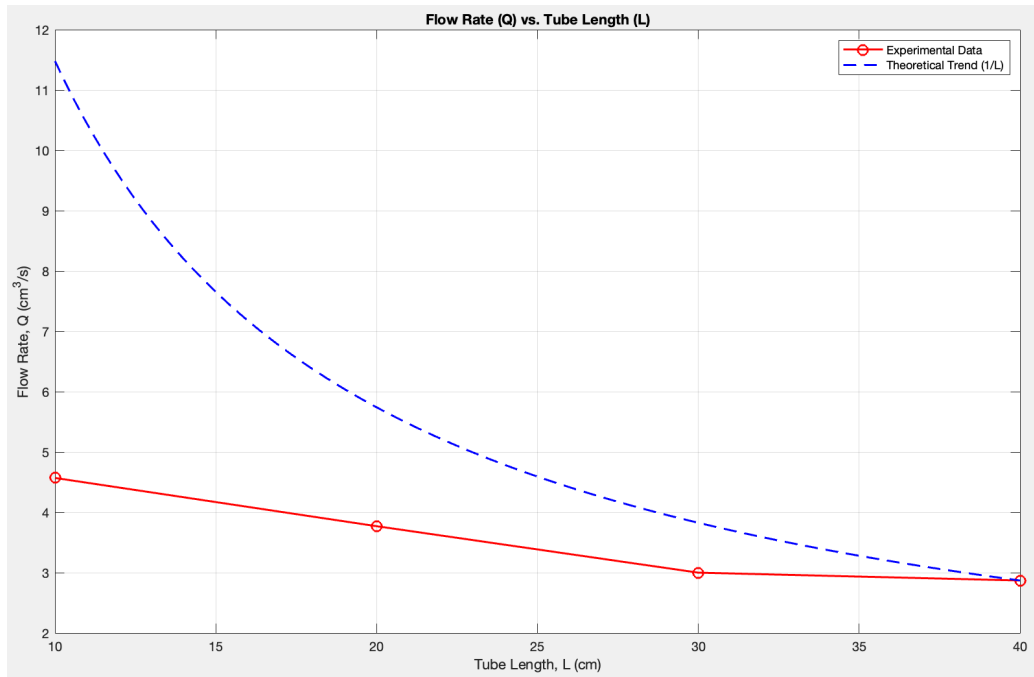


Figure 3: Flow Rate vs. Length of the Tube

For both graphs, we cannot say the measured values satisfy the theoretical values because we can obviously see the data points are outliers. This is because of our experimental error, which is measured 40% before.

This error and the outliers are not unexpected because our experimental setup and measurements have some problems and not precise so we cannot expect lower errors.

However, despite our setup, the error and shapes of the graphs are not too bad.